FEEDBACK CONTROL SYSTEMS LAB

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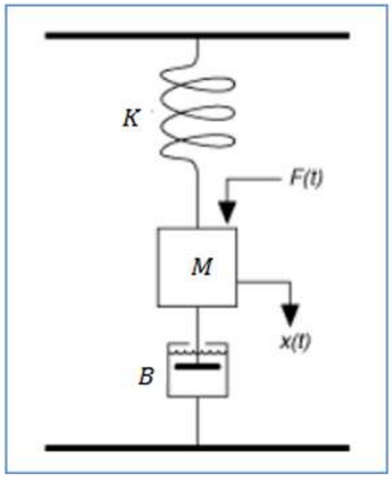
Question: 01



Given a mass-spring damper system

B=damping constant

M=mass

K=spring constant

F=u=force

The state space model for this system is shown below

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}0 & 1\\ -\frac{k}{m} & -\frac{c}{m}\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}0\\ \frac{1}{m}\end{bmatrix}u$$

$$y = \begin{bmatrix}1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix}$$

a) Define the state-space model above using the ss function in MATLAB.
   Set c=1, m=1, k=50.

b) Use MATLAB to simulate the result. Explain each step in detail.

c) Find the transfer function from the state-space model.

d) Find the responses of the system using SIMULINK and insert the screen shots.

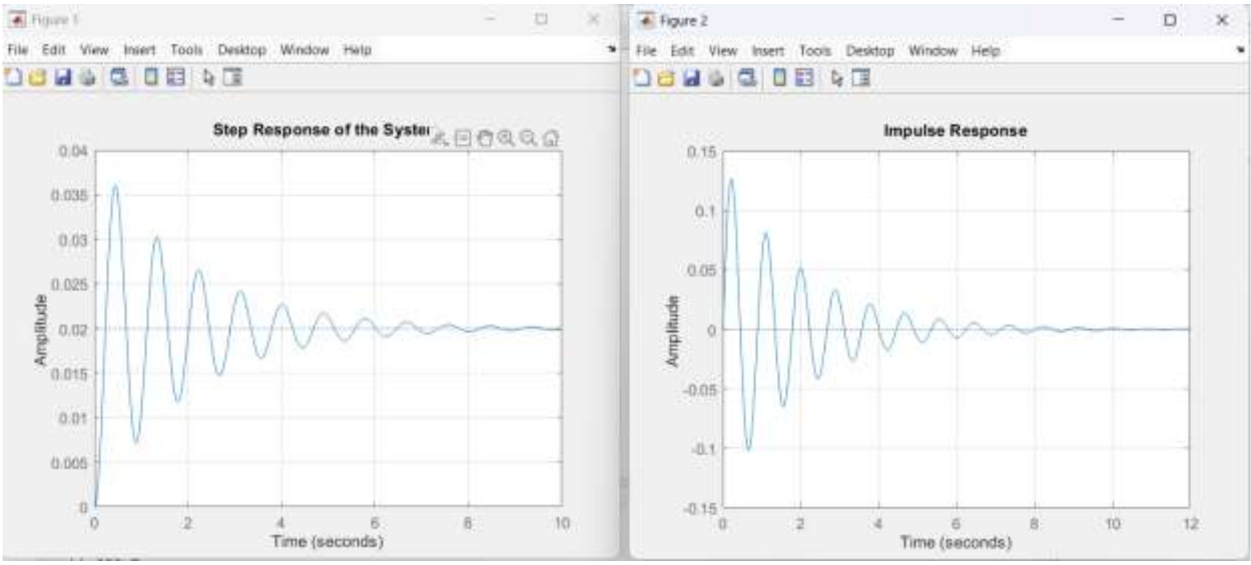
untitled.m *

```matlab
% a)
A = [0 1; -50 -1]; %A = [0 1; -k/m -c/m];
B = [0; 1];        %B = [0; 1/m];
C = [1 0];
D = 0;

sys = ss(A, B, C, D);
disp('State-Space Model:');
sys

% b)
t = 0:0.01:10;
figure;
step(sys, t); % step response
grid on;
title('Step Response of the System');

% impluse
figure;
impulse(sys);
title('Impulse Response');
grid on;
% c)
tf_sys = tf(sys)
disp('Transfer Function:');
tf_sys
```

Command Window

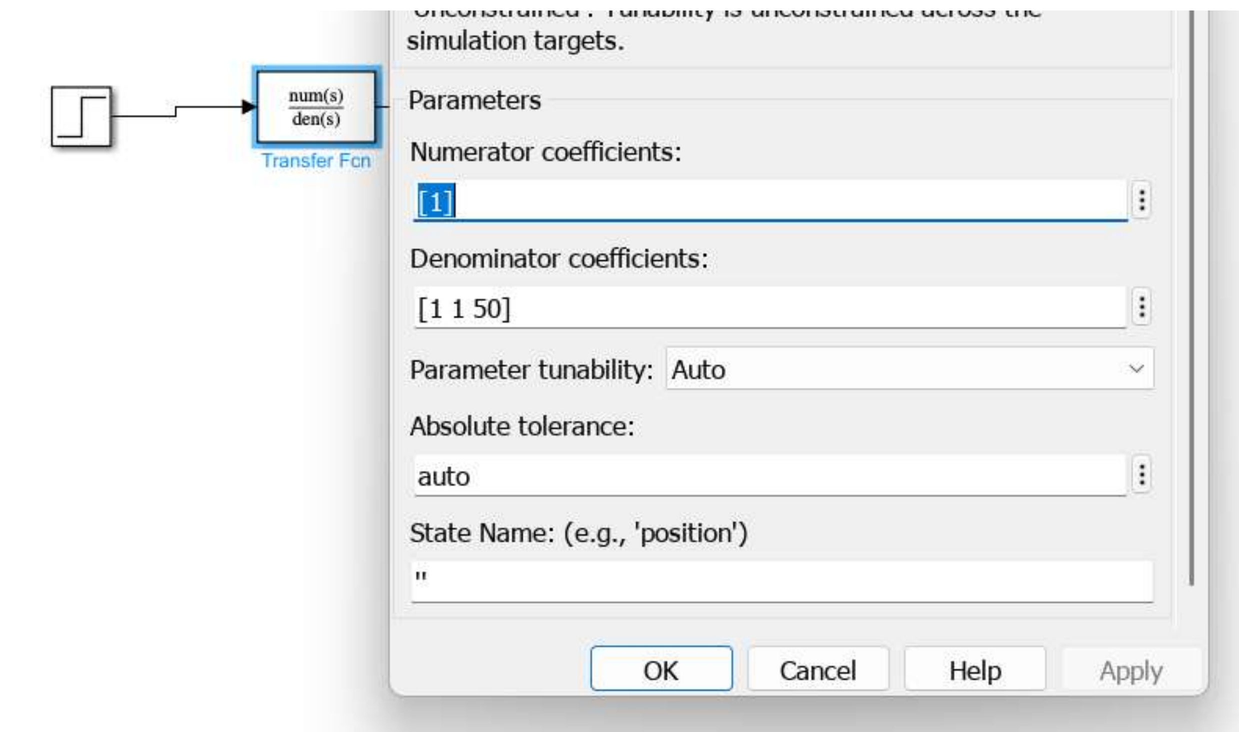
```
State-Space Model:

sys =

  A =
        x1   x2
   x1    0    1
   x2  -50   -1

  B =
       u1
   x1   0
   x2   1

  C =
       x1  x2
   y1   1   0

  D =
       u1
   y1   0
```

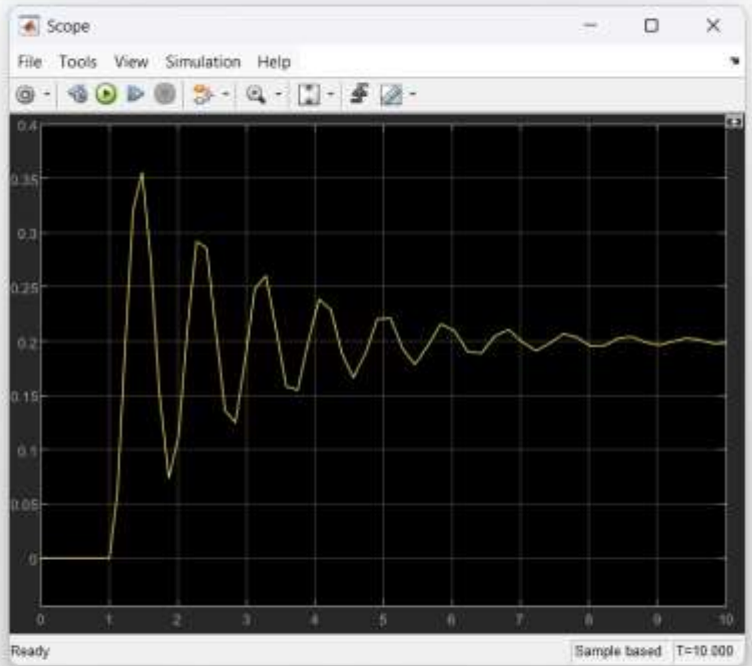
```
tf_sys =

        1
  ------------
  s^2 + s + 50
```



Simulink

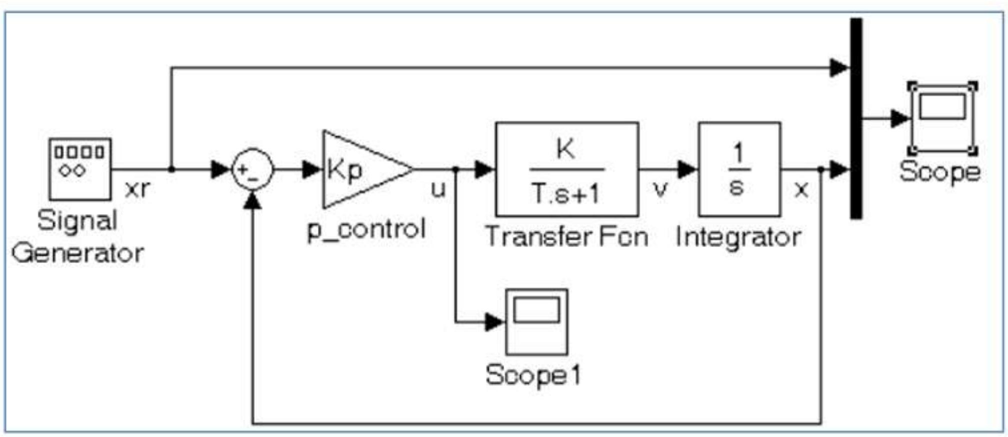
Edit properties of transfer function





Question: 02

Create a SIMULINK model with a first order system, with gain, K =1, and time constant, T = 0.1 sec. Simulate a square wave input with unit amplitude and frequency of 0.3 Hz. The sample time is 0.001 sec. View the reference position, xr(t), input, $u(t)$, and actual position, $x(t)$, through a scope, as in Figure 3.13 below. Experiment with different values of Kp and observe how the system response changes. Plot the results.



From Library Browser, drag:

Sources:

- **Signal Generator**

Math Operations:

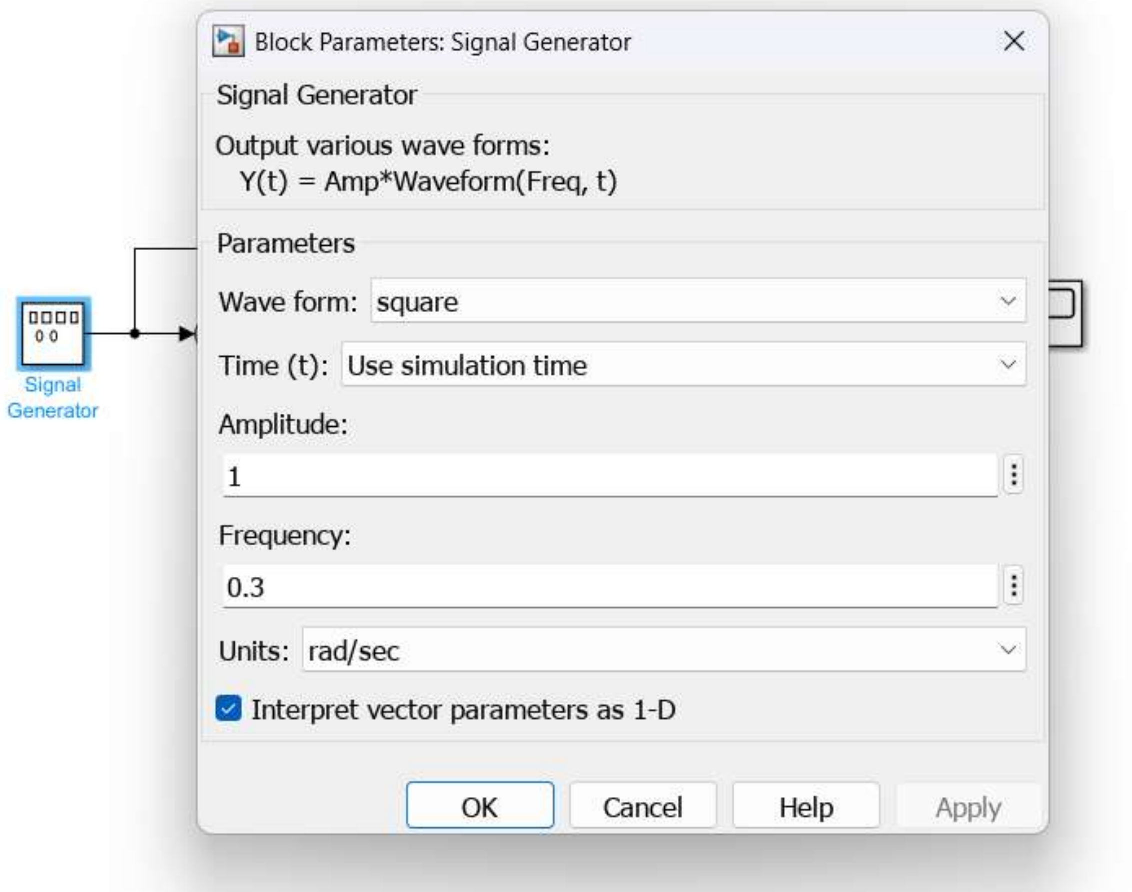
- **Sum**
- **Gain** (for Kp)

Continuous:

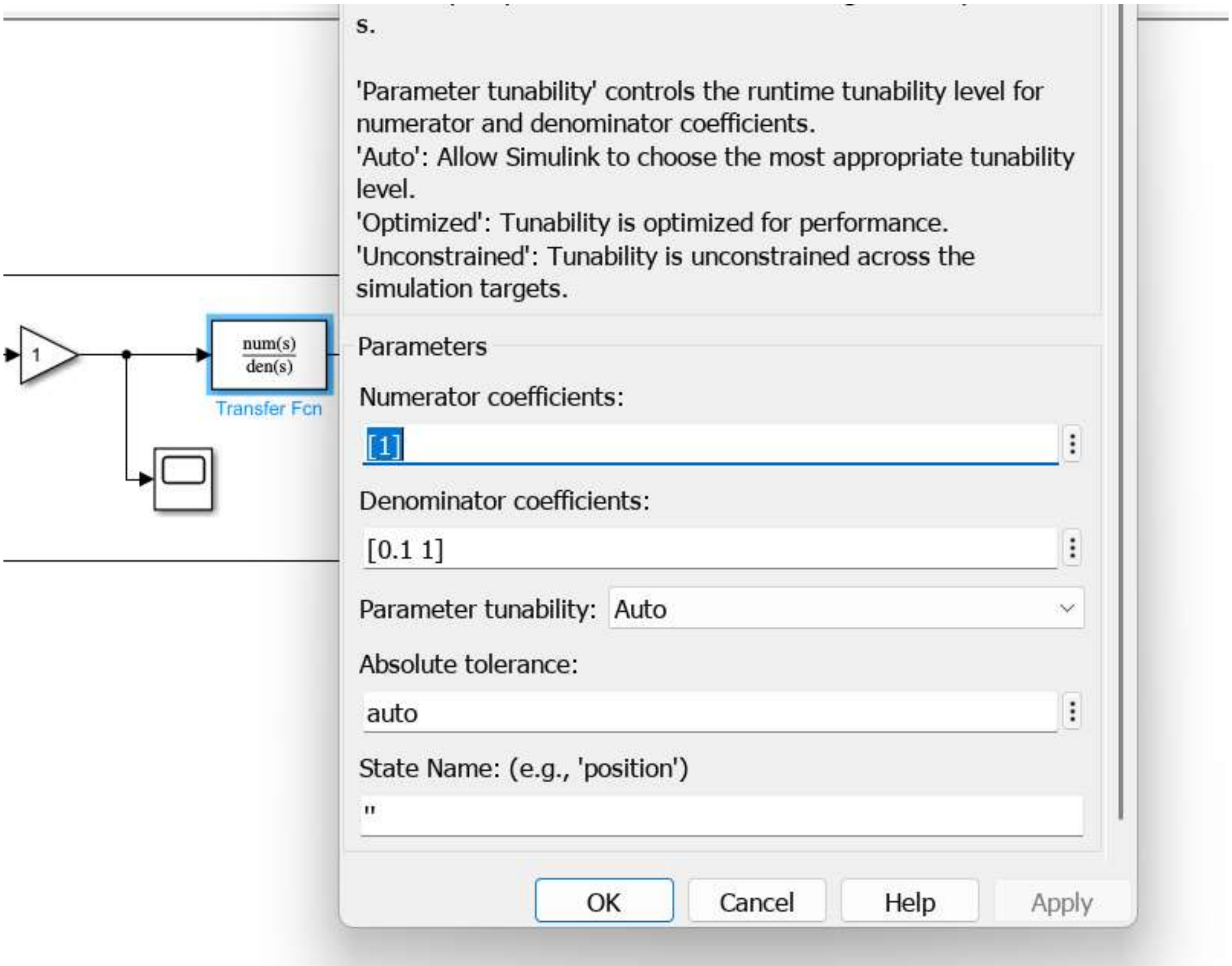
- **Transfer Fcn**
- **Integrator**
- **Mux**

Sinks:

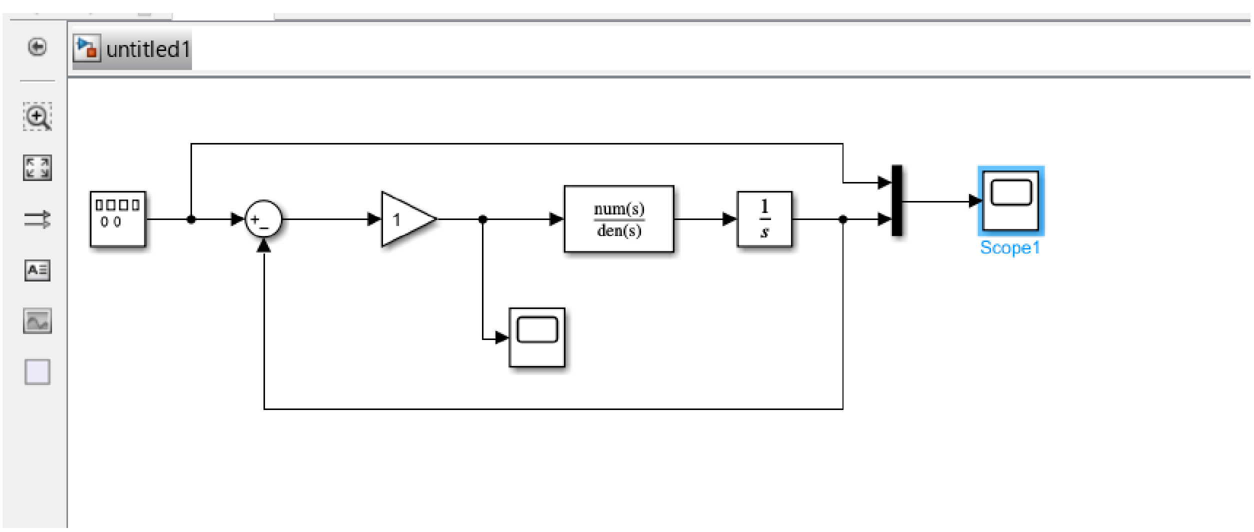
- **Scope** (2 scopes)

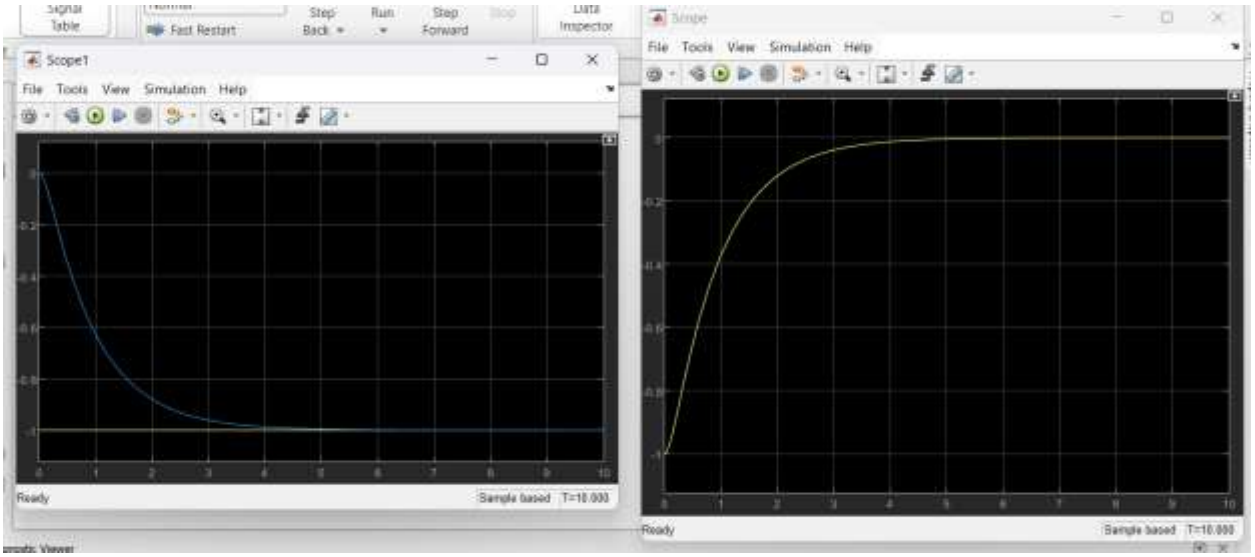


Set transfer function num and den

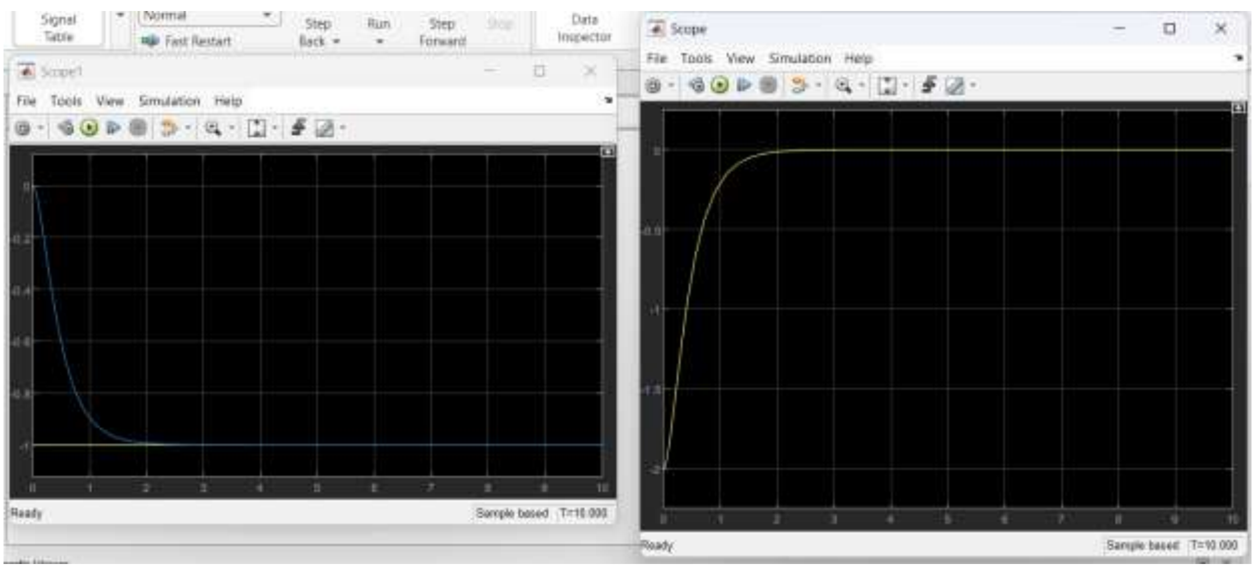


K=1

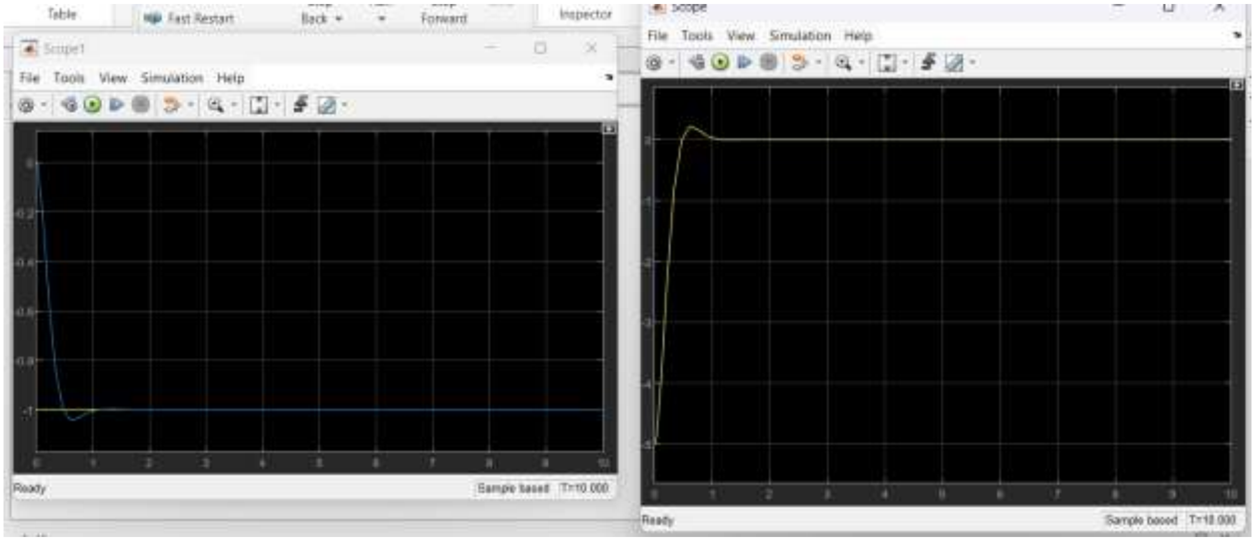



Scope1
File Tools View Simulation Help
Ready
Sample based
T=10.000
File Tools View Simulation Help
Ready
Sample based
T=10.000

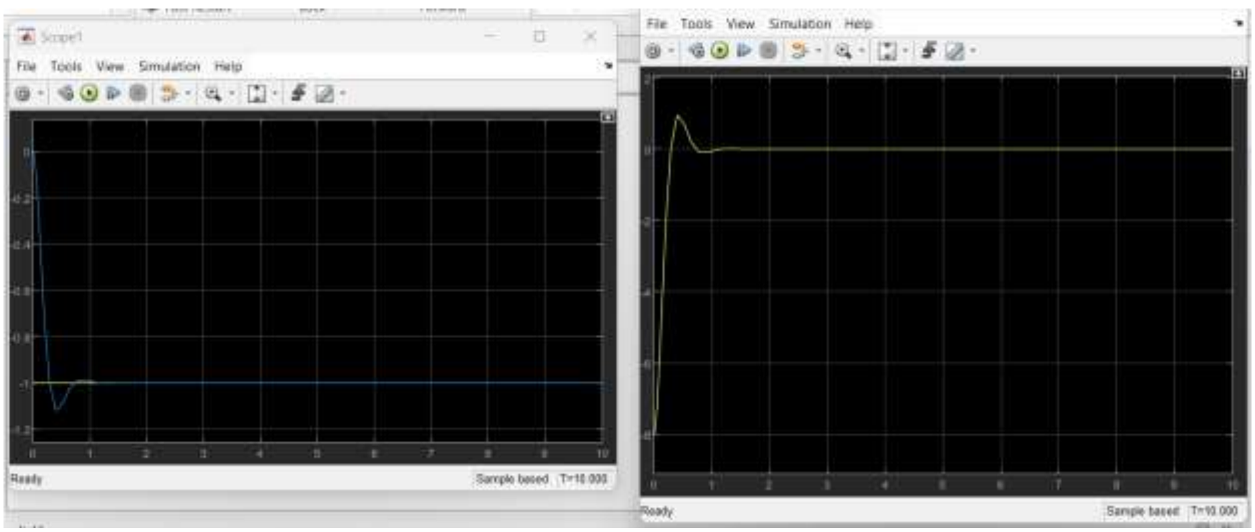
K=2


Signal Table
Fast Restart
Step Back
Run
Step Forward
Data Inspector
Scope
File Tools View Simulation Help
Ready
Sample based
T=10.000
File Tools View Simulation Help
Ready
Sample based
T=10.000

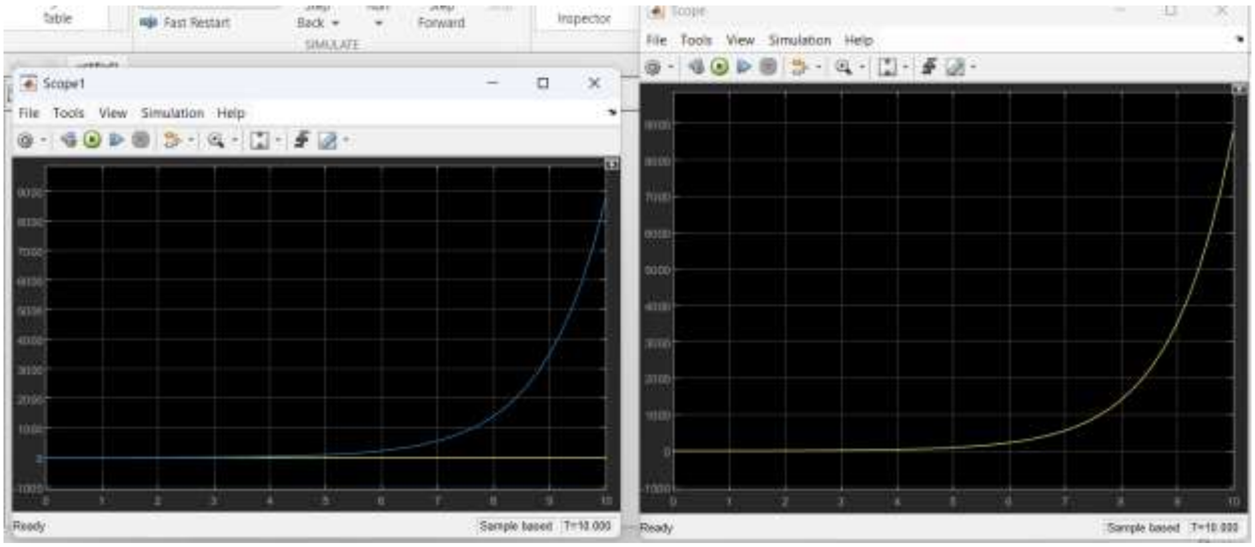
K=5


Scope1
File Tools View Simulation Help
Ready
Sample based
T=10.000
Scope
File Tools View Simulation Help
Ready
Sample based
T=10.000

K=8


Scope1
File Tools View Simulation Help
Ready
Sample based
T=10.000
File Tools View Simulation Help
Ready
Sample based
T=10.000

K= -1



he proportional gain Kp significantly affects the performance of the control system.

- For **Kp=−1**: The system becomes unstable due to positive feedback, and the output diverges.
- For **Kp=1**: The system is stable but responds slowly with noticeable steady-state error.
- For **Kp=2**: The response becomes faster and the steady-state error decreases.
- For **Kp=5**: The system shows a fast response with very small steady-state error, but slight overshoot may occur.
- For **Kp=8**: The response is very fast; however, the system may exhibit oscillations and reduced stability.

Conclusion:
Increasing Kp improves response speed and reduces steady-state error, but excessively high values can lead to oscillations and instability.